

A DEV ENVIRONMENT SETUP

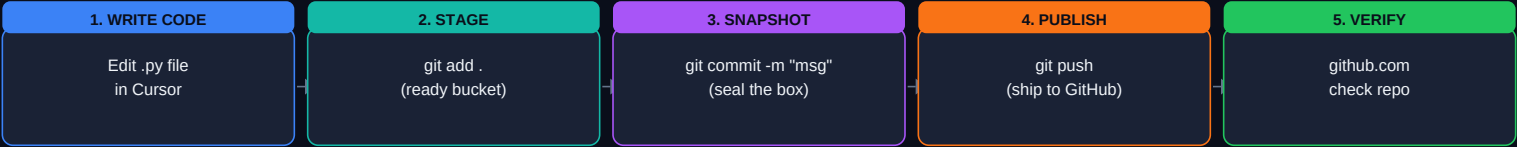
Tools Installed

- Python 3 (via Homebrew)
- Cursor (AI code editor)
- Git 2.50 (version control)
- GitHub account (portfolio host)

Key Commands

`python3 file.py` → run a Python script
`ls` → list files in folder
`cd folder` → navigate into folder
`git --version` → check Git is installed

B GIT & GITHUB WORKFLOW



KEY CONCEPTS:

C PYTHON BASICS

Staging area — the 'ready bucket' before committing **Commit** — a permanent snapshot with a label (-m flag = message) **Push** — sends local snapshots to GitHub cloud

What You Learned

- `print()` sends output to the terminal
- `python3 <file>` runs your script
- Save file (Cmd+S) before running
- Terminal shows output between \$ prompts
- Functions take inputs, produce outputs

Your First Script (Hello.py)

```
print("Hello, World!")
print(2+2)
```

→ output: Hello, World! and 4

D THE AI FAMILY TREE — Concentric layers

1

ARTIFICIAL INTELLIGENCE
Any technique making machines appear intelligent — including simple rules.
EXAMPLE Spam filter: IF email contains 'win a prize' → mark spam

2

MACHINE LEARNING
Machine learns rules from data, not hand-coded. Shows examples → finds patterns.
EXAMPLE Spam filter trains on 1M emails — no rules written by humans

3

DEEP LEARNING
ML using layered neural networks. Powerful for text, images, audio. Often opaque.
EXAMPLE Face unlock on iPhone; document recognition; speech-to-text

4

GENERATIVE AI
Deep learning that creates new content — text, code, images, audio.
EXAMPLE Claude writing a summary; GPT generating code; Midjourney creating images

5

AGENTS
GenAI that takes actions, uses tools, plans across steps toward a goal.
EXAMPLE AI that researches competitors, reads pages, synthesises & delivers a report

E WHAT SITS PARALLEL AT EACH LAYER

■ AI

- Rule-based / Expert Systems
- Search & Optimisation (A*)
- Knowledge Graphs
- RPA (Robotic Process Automatio

■ ML

- Classical / Statistical ML
- Probabilistic / Bayesian
- Ensemble Methods (XGBoost)

■ Deep Learning

- Discriminative Models (CNNs)
- Deep Reinforcement Learning
- Representation Learning

■ Generative AI

- Standalone LLMs (no tools)
- RAG Systems (retrieval only)
- Fine-tuned Domain Models
- Multimodal Models

■ Agents

- Copilot / Assistant
- RAG + Tools
- Multi-Agent Systems

F THE CONSULTING LENS — From client problem → AI branch

What the client says	AI archetype	Technical approach
"Automate document review"	Document Intelligence	Supervised ML + LLM extraction
"Reduce fraud losses"	Prediction + Anomaly	Supervised / Unsupervised ML
"Chatbot for staff queries"	Knowledge Assistant	RAG + LLM (Generative AI)
"Personalise recommendations"	Recommendation Engine	Collaborative Filtering / ML
"Monitor our operations"	Monitoring Agent	Agents + Anomaly Detection
"Optimise supply chain"	Optimisation	Operations Research + ML